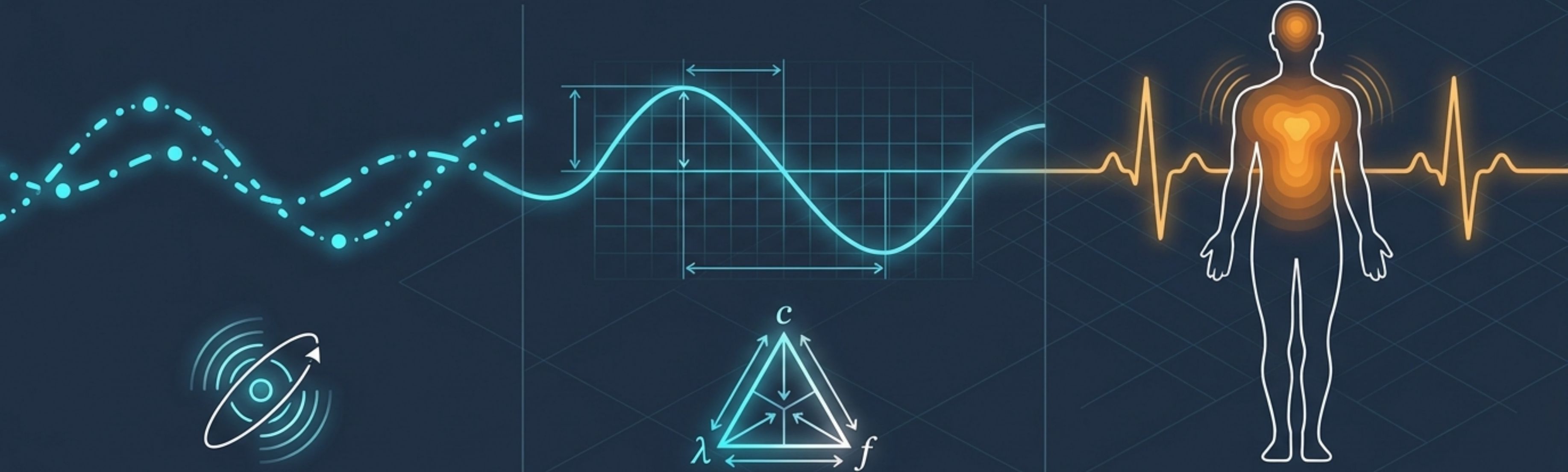


The Blueprint of the Invisible

Electromagnetic Waves, Mathematical Principles, and Biological Safety



From Abstract Physics to Physical Impact



Pillar 1: The Phenomenon

The birth and nature of the electromagnetic wave.

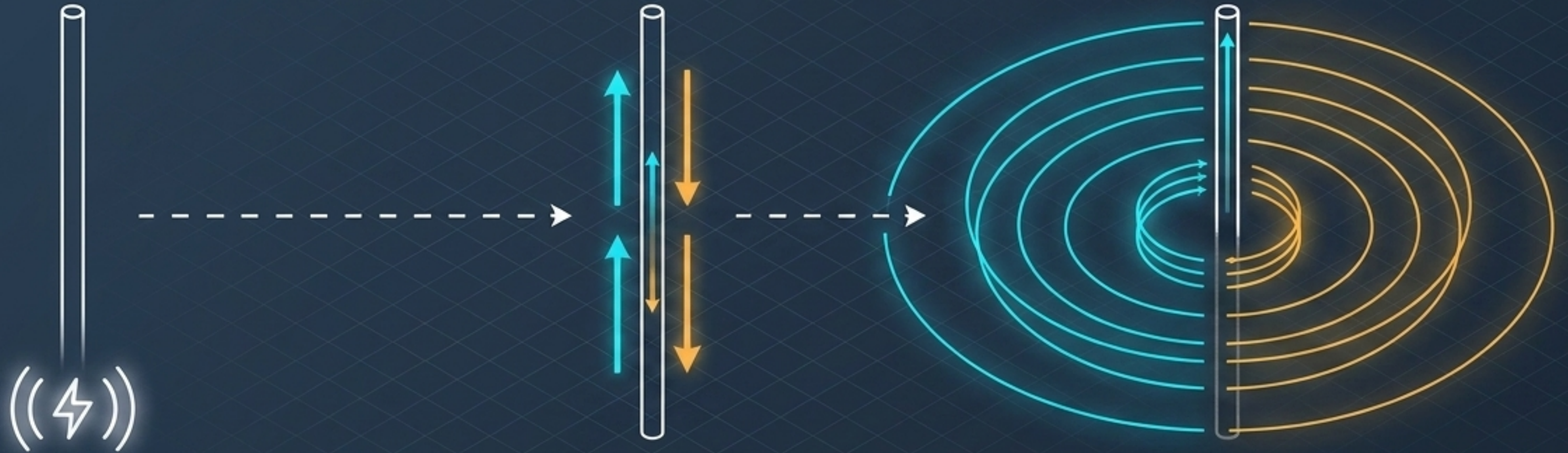
Pillar 2: The Universal Law

The mathematical relationship between frequency, wavelength, and the speed of light.

Pillar 3: The Biological Interaction

How invisible radio frequencies interact with, heat, and resonate within human tissue.

The Birth of an Electromagnetic Wave



Step 1: Application (The Source)

Alternating current (AC) or voltage is applied to a conductor (an antenna).

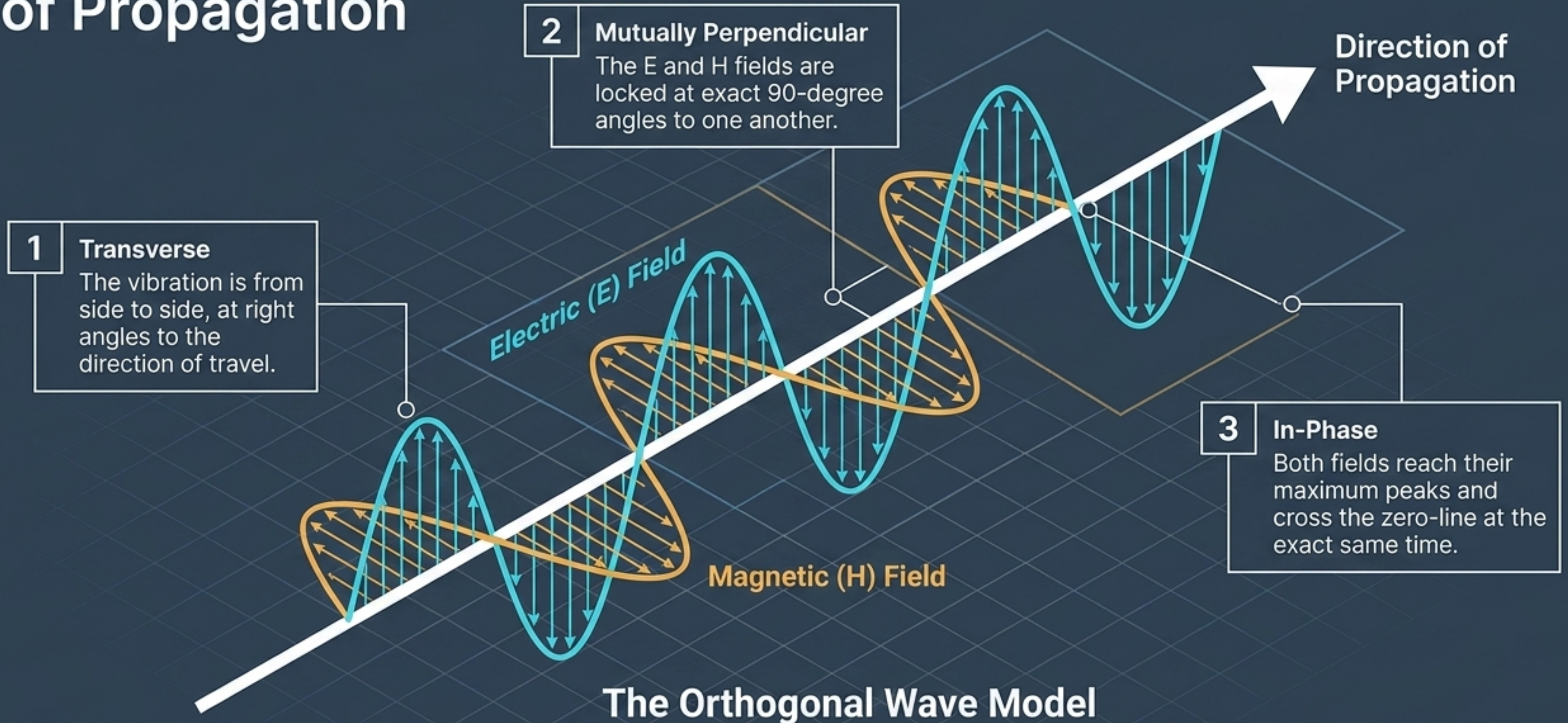
Step 2: Oscillation (The Movement)

Electrons within the wire vibrate back and forth in synchronization with the applied current.

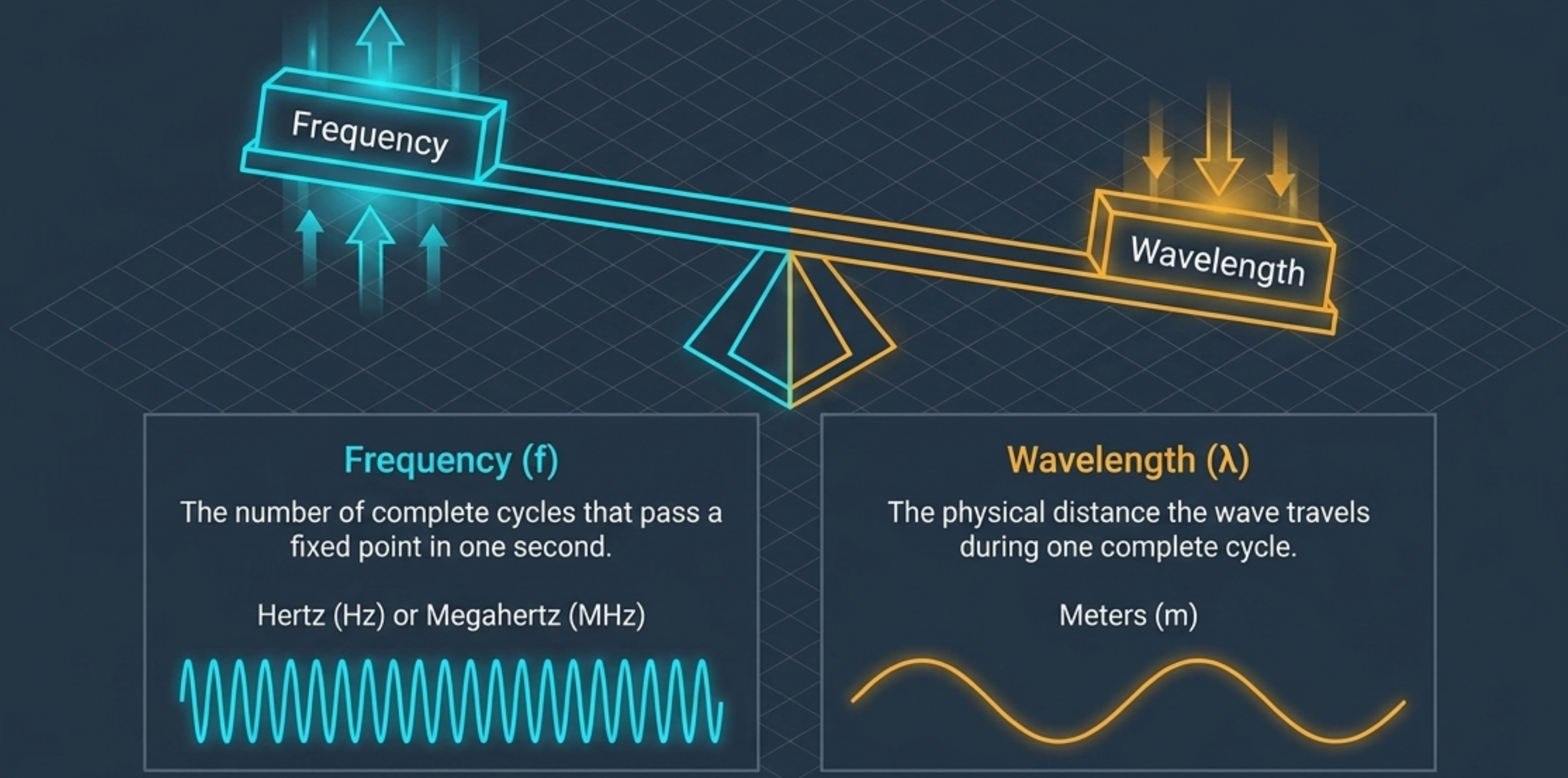
Step 3: Radiation (The Detachment)

The moving electrons generate expanding electric and magnetic fields that detach and radiate into space at 300,000,000 meters per second.

The Anatomy of Propagation



The Inverse Relationship



The Principle: Because the speed of light is constant, as frequency increases, wavelength must proportionally decrease.

The Velocity Constant: Calculating the Wave

$$f \times \lambda = c \text{ (Speed of Light)}$$

To find **Frequency**:

$$f = \frac{300}{\lambda}$$

300

f (MHz)

λ (meters)

To find **Wavelength**:

$$\lambda = \frac{300}{f}$$



Requirement: To use this simplified 300 rule, frequency must be in Megahertz and wavelength must be in meters.

Applying the Universal Law

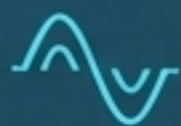
Case Study 1: Finding Frequency

The 80-Meter Band

Given: Wavelength = 80 meters

Calculation: $f = 300 / 80$

Result: $f = 3.75$ MHz



Context: Falls directly into the High Frequency (HF) spectrum.

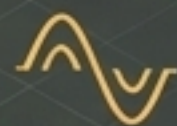
Case Study 2: Finding Wavelength

The 2-Meter Band

Given: Frequency = 146.5 MHz

Calculation: $\lambda = 300 / 146.5$

Result: $\lambda = 2.05$ meters



Context: Falls directly into the Very High Frequency (VHF) spectrum.

Radio Waves vs. Radioactivity



Ionizing Radiation

Examples

X-Rays, Gamma Rays, Nuclear Particles

Mechanism

Carries enough energy to strip electrons from atoms.

Biological Risk

Causes cellular and genetic damage (DNA disruption).



Non-Ionizing Radiation (RF Energy)

Examples

AM/FM Radio, Wi-Fi, Amateur Radio, Cell Phones

Mechanism

Moves electrons but cannot break chemical bonds.

Biological Risk

Primary risk is tissue heating (thermal effects) and nerve stimulation at lower frequencies. Does not cause genetic damage.

Health Canada's Safety Code 6

Core Purpose

Establishes strict safety limits for human exposure to Radiofrequency (RF) electromagnetic energy to prevent adverse thermal effects.

Scope

Covers frequencies ranging from 3 kHz up to 300 GHz.

The Standard of Measurement: SAR

Specific Absorption Rate (SAR) measures the rate at which RF energy is deposited into body tissue per unit mass.

- Unit: Watts per kilogram (W/kg).
- The Rule: RF energy absorption follows a thermal time constant of 6 minutes. Time-averaged exposure must not exceed specific biological limits.



Evaluating the Environment



Controlled Environments

Areas where individuals are aware of the RF exposure, understand the health risks, and can use mitigation strategies.

0.4 W/kg

Whole-Body SAR Limit

Safety Factor: Incorporates a 10x safety margin below the scientifically established hazard threshold (4 W/kg).



Uncontrolled Environments

Areas accessible to the general public, who have no awareness training and no means to assess or mitigate their exposure.

0.08 W/kg

Whole-Body SAR Limit

Safety Factor: Incorporates a stricter 50x safety margin to protect vulnerable populations of all ages and body sizes.

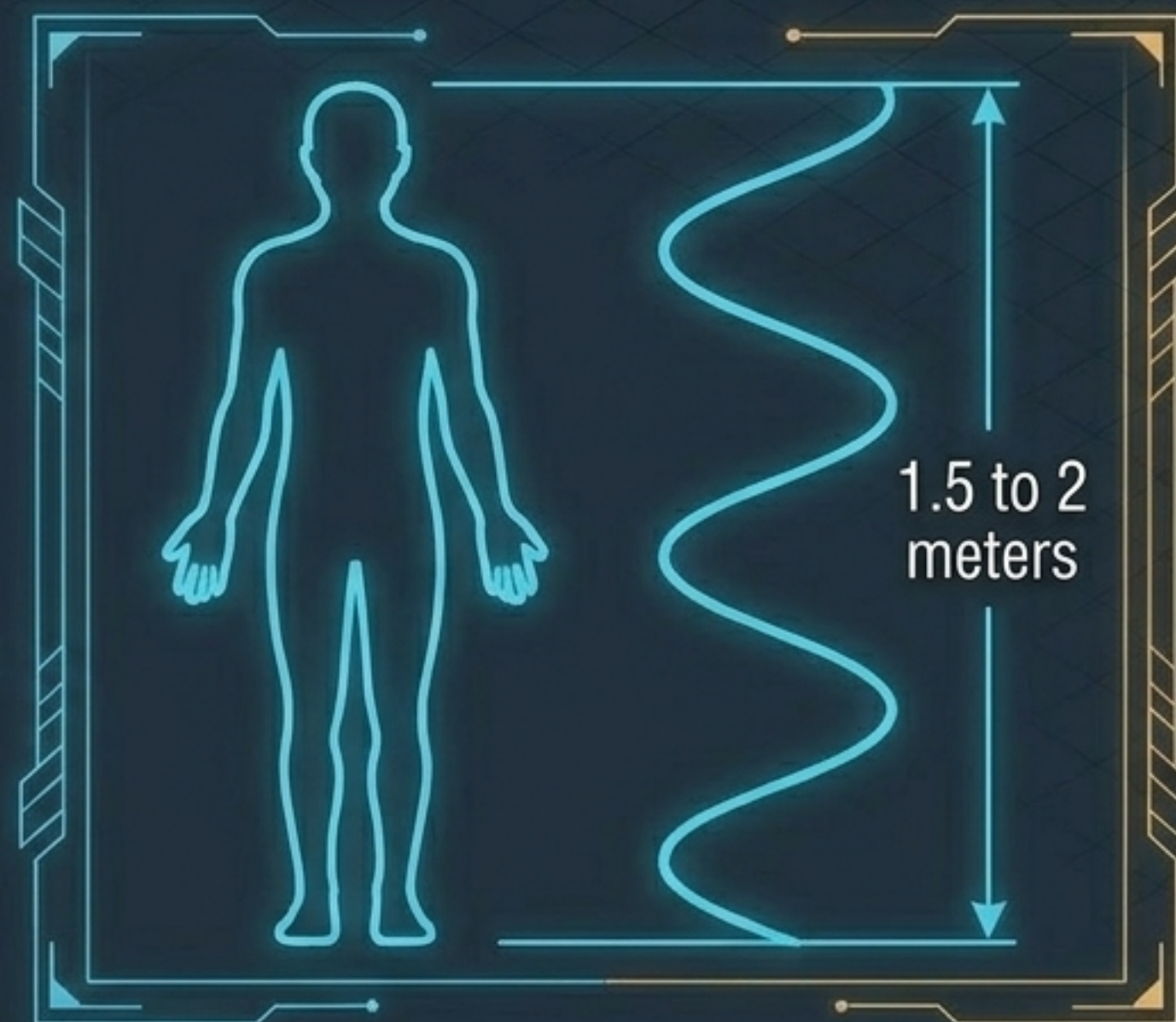
The Danger of Biological Resonance

The Insight: Safety Code 6 identifies the 30 MHz to 300 MHz range (VHF) as posing the greatest risk to human tissue. Why?

The Math Connection

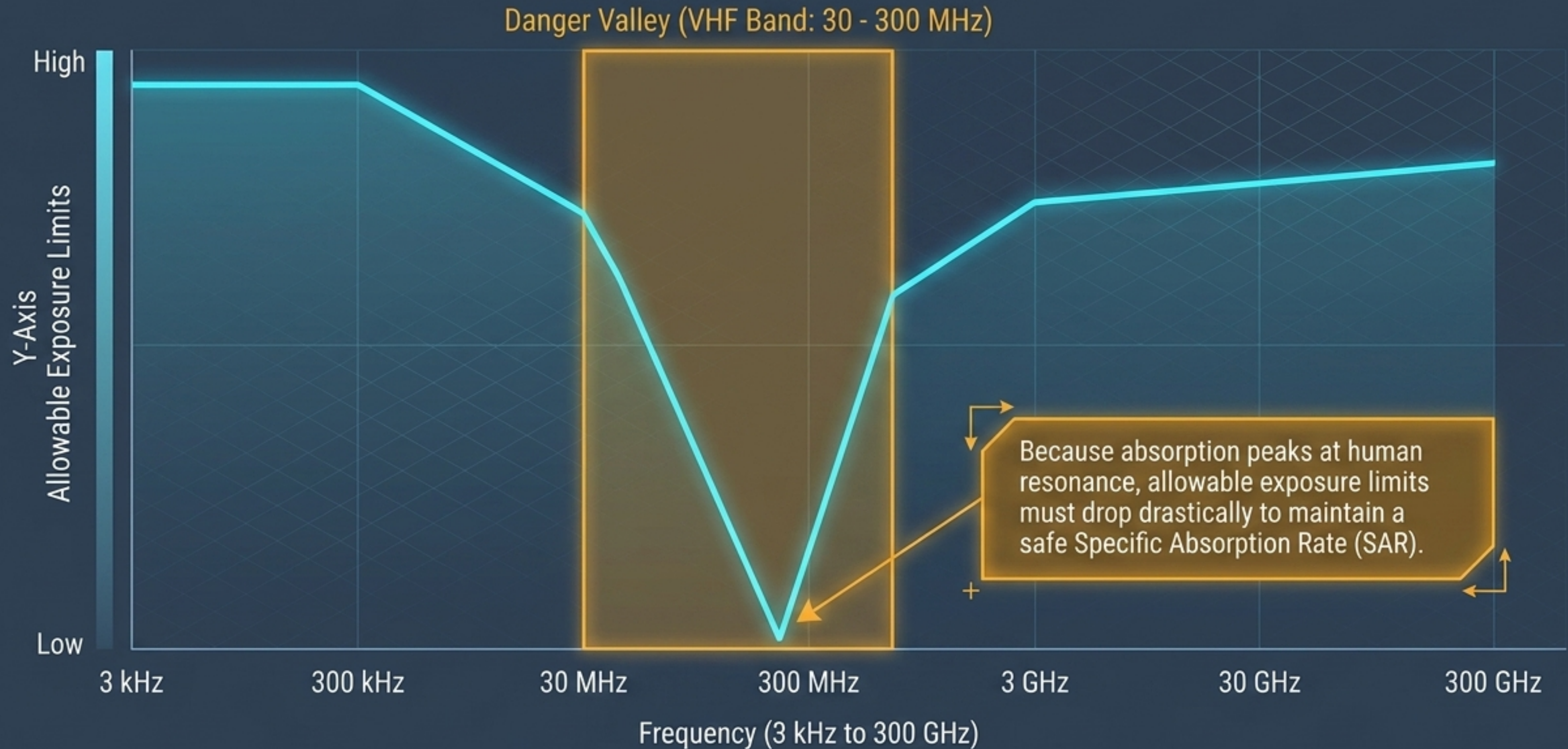
$$\lambda = \frac{300}{300 \text{ MHz}} = 1 \text{ meter}$$

$$\lambda = \frac{300}{30 \text{ MHz}} = 10 \text{ meters}$$



The Conclusion: The physical wavelengths of this spectrum directly match the physical heights of human beings. In this frequency range, the human body acts as a highly efficient, resonant receiving antenna, absorbing energy at maximum capacity.

The Risk Spectrum Heat Map



Operational RF Mitigation



Mastering the Invisible

